

The Constants of Nature

 E h : K = ℏ<!-- ℏ --> m e c 2 (t p m p m p 16 C 0 −<!-- − --> 4 ⋈<!-- ⋈ --> g 9 ⊠<!-- ⊠ -->) 	 m de = 3 s ⋈<!-- ⋈ --> 3 3 4 (16 4 π<!-- π -->) (m n + m e) (1 + l m (ω<!-- ω --> i) (l 2 14 ⋈<!-- ⋈ --> r 3 ⊠<!-- ⊠ -->) 	 ℰ<!-- ℰ --> p = joule GeV (l p 2 m p t p 2) (1 −<!-- − --> 1 2 R e (i ℓ<!-- ℓ --> ′) ⋈<!-- ⋈ --> 3 3 4 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> R = ℏ<!-- ℏ --> ⋈<!-- ⋈ --> 3 4 (t p q p l p m e) (1 + √<!-- √ --> 4 D 00 ⋈<!-- ⋈ --> 1 3 3 4 ⋈<!-- ⋈ --> 4 ⊠<!-- ⊠ -->) 	 c = (l p t p) (1 + √<!-- √ --> G C 1 (⋈<!-- ⋈ --> 1 −<!-- − --> ⋈<!-- ⋈ --> 2) ⊠<!-- ⊠ -->) 	 F 90 = farad (1 + C d 6 ⋈<!-- ⋈ --> 1 + ⋈<!-- ⋈ --> 3 + ⋈<!-- ⋈ --> 4) ⊠<!-- ⊠ -->) 	 K J = ℏ<!-- ℏ --> ⋈<!-- ⋈ --> 1 π<!-- π --> (t p q p l p m p) (1 −<!-- − --> 7 8 (8 4 π<!-- π -->) 2 ⋈<!-- ⋈ --> 1 2 + ⋈<!-- ⋈ --> 2 2) ⊠<!-- ⊠ -->) 	 g de = 2 π<!-- π --> 8 K (1 + 16 P up ⋈<!-- ⋈ --> 1 + ⋈<!-- ⋈ --> 3 + ⋈<!-- ⋈ --> 4 + ⋈<!-- ⋈ --> 4 2) ⊠<!-- ⊠ -->) 	 t p = 5.39125836832313 …<!-- … --> ×<!-- × --> 10 −<!-- − --> 44 s l p = 1.61625918175645 …<!-- … --> ×<!-- × --> 10 −<!-- − --> 35 m q p = 1.87554596713962 …<!-- … --> ×<!-- × --> 10 −<!-- − --> 18 C T p = 1.4167869859795 …<!-- … --> ×<!-- × --> 10 32 K m p = 2.17642683817579 …<!-- … --> ×<!-- × --> 10 −<!-- − --> 8 kg 	 ⊠<!-- ⊠ --> = 0.99999919973626 …<!-- … --> 10 −<!-- − --> 7 5 7 hyperbolic inversion boundary number of dimensions break in scale symmetry																		
 Δ<!-- Δ --> c 5 = E i 2 π<!-- π --> (t p l p m p) (1 −<!-- − --> √<!-- √ --> ζ<!-- ζ --> (2) ⋈<!-- ⋈ --> g 9 2 ⊠<!-- ⊠ -->) 	 m p = 1 ⋈<!-- ⋈ --> 3 3 4 √<!-- √ --> 16 4 π<!-- π --> 2 (m n + m e) (1 + 16 2 π<!-- π -->) ⋈<!-- ⋈ --> r 3 ⊠<!-- ⊠ -->) 	 R E = 2 π<!-- π --> ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 (l p 2 m p t p 2) (1 + 1 2 R e (i ℓ<!-- ℓ --> ′) ⋈<!-- ⋈ --> 3 3 4 ⊠<!-- ⊠ -->) 	 Z 0 = 4 π<!-- π --> (l p 2 m p t p q p) (1 + 1 2 R e (i ℓ<!-- ℓ --> ′) ⋈<!-- ⋈ --> 3 3 4 ⊠<!-- ⊠ -->) 	 c = ℏ<!-- ℏ --> ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 (t p q p l p m e) (1 + √<!-- √ --> 6 C R 1 (⋈<!-- ⋈ --> 1 −<!-- − --> ⋈<!-- ⋈ --> 2) ⊠<!-- ⊠ -->) 	 a 0 = 1 ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 (l p m p t p 2) (1 −<!-- − --> √<!-- √ --> 6 C R 1 (⋈<!-- ⋈ --> 1 −<!-- − --> ⋈<!-- ⋈ --> 2) ⊠<!-- ⊠ -->) 	 Φ<!-- Φ --> 0 = π<!-- π --> ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 (l p m p t p q p) (1 + 7 8 (8 4 π<!-- π -->) 2 ⋈<!-- ⋈ --> 1 2 + ⋈<!-- ⋈ --> 2 2) ⊠<!-- ⊠ -->) 	 h e = coulomb e (l p 2 m p t p q p) (1 + 7 8 (8 4 π<!-- π -->) 2 ⋈<!-- ⋈ --> 1 2 + ⋈<!-- ⋈ --> 2 2) ⊠<!-- ⊠ -->) 	 b ′ = V peak 2 π<!-- π --> (1 t p T p) (1 + 18 16 (2 π<!-- π --> s) 2 ⋈<!-- ⋈ --> 1 2 −<!-- − --> ⋈<!-- ⋈ --> 3 2 −<!-- − --> ⋈<!-- ⋈ --> 4 2) ⊠<!-- ⊠ -->) 	 K J = kelvin second (1 t p T p) (1 + 18 16 (2 π<!-- π --> s) 2 ⋈<!-- ⋈ --> 1 2 −<!-- − --> ⋈<!-- ⋈ --> 3 2 −<!-- − --> ⋈<!-- ⋈ --> 4 2) ⊠<!-- ⊠ -->) 	 G V i = 1.01494160640965 …<!-- … --> K = 2.02988321281930 …<!-- … --> C d = 0.29156090430818 …<!-- … --> D d = 1.79162281206959 …<!-- … --> ω<!-- ω --> 1 = 0.764977018528596 …<!-- … --> + 1.32497062714087 …<!-- … --> i ω<!-- ω --> 2 = 1.529954037057192 …<!-- … --> 1st hyperbolic partition constant 2nd hyperbolic partition constant 3rd hpc 4th hpc hyperbolic radius constant hyperbolic radian constant																	
 ℏ<!-- ℏ --> : Hz = joule 2 π<!-- π --> (l p m p t p 2) (1 −<!-- − --> √<!-- √ --> ζ<!-- ζ --> (2) ⋈<!-- ⋈ --> g 9 2 ⊠<!-- ⊠ -->) 	 G ℏ<!-- ℏ --> c = GeV (t p 2 l p 2 m p) 2 (1 −<!-- − --> 4 π<!-- π -->) ⋈<!-- ⋈ --> r 3 ⊠<!-- ⊠ -->) G = (l p 3 m p t p) (1 −<!-- − --> 4 π<!-- π -->) ⋈<!-- ⋈ --> r 3 ⊠<!-- ⊠ -->) 	 ℏ<!-- ℏ --> : 1 m = 1 second (t p t p) (1 + 1 2 I m (i ℓ<!-- ℓ --> ′) ⋈<!-- ⋈ --> 3 3 4 ⊠<!-- ⊠ -->) 	 A vel = ⋈<!-- ⋈ --> 1 2 (l p t p) (1 −<!-- − --> 1 2 I m (i ℓ<!-- ℓ --> ′) ⋈<!-- ⋈ --> 3 3 4 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = m s (m p) (1 −<!-- − --> 18 6 π<!-- π -->) ⋈<!-- ⋈ --> 1 3 3 4 ⋈<!-- ⋈ --> 4 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s ′ = 2 ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 S (1 −<!-- − --> 18 (6 π<!-- π -->) ⋈<!-- ⋈ --> 1 3 3 4 ⋈<!-- ⋈ --> 4 ⊠<!-- ⊠ -->) 	 A mom = ⋈<!-- ⋈ --> 1 2 (l p m p t p) (1 −<!-- − --> 1 2 I m (i ℓ<!-- ℓ --> ′) ⋈<!-- ⋈ --> 3 3 4 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s ′ = 2 ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 S (1 −<!-- − --> 18 (6 π<!-- π -->) ⋈<!-- ⋈ --> 1 3 3 4 ⋈<!-- ⋈ --> 4 ⊠<!-- ⊠ -->) 	 K : Hz = kelvin second (1 t p T p) (1 + 18 16 (2 π<!-- π --> s) 2 ⋈<!-- ⋈ --> 1 2 −<!-- − --> ⋈<!-- ⋈ --> 3 2 −<!-- − --> ⋈<!-- ⋈ --> 4 2) ⊠<!-- ⊠ -->) 	 ℏ<!-- ℓ --> : K = 2 π<!-- π --> second (t p T p) (1 −<!-- − --> 18 16 (2 π<!-- π --> s) 2 ⋈<!-- ⋈ --> 1 2 −<!-- − --> ⋈<!-- ⋈ --> 3 2 −<!-- − --> ⋈<!-- ⋈ --> 4 2) ⊠<!-- ⊠ -->) 	 G CF = 0.697774657964819 …<!-- … --> C R 1 = 0.998136044598509 …<!-- … --> F FF = 1.22674021072035 …<!-- … --> M = −<!-- − --> 1.74756459463318 …<!-- … --> figure eight knot hyperbolic volume Catalan's constant domino tiling constant dimer constant omega_1 constant omega_2 constant																	
 h = (l p 2 m p t p) (1 + √<!-- √ --> ζ<!-- ζ --> (2) ⋈<!-- ⋈ --> g 9 2 ⊠<!-- ⊠ -->) 	 m tri = ⋈<!-- ⋈ --> 1 −<!-- − --> 5 18 kilogram √<!-- √ --> 4 s 2 π<!-- π --> (m p 4) (1 −<!-- − --> 3 ω<!-- ω --> 1 2 ⋈<!-- ⋈ --> r 3 2 ⊠<!-- ⊠ -->) 	 K cl = ⋈<!-- ⋈ --> 1 −<!-- − --> 7 f 1 √<!-- √ --> 4 s (t p 2 l p 2 A mass) (1 + 3 ω<!-- ω --> 2 2 ⋈<!-- ⋈ --> r 3 2 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> de = −<!-- − --> C CF F (2 s 6 π<!-- π -->) (m n m de) (1 −<!-- − --> ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 (35 L 2) ⋈<!-- ⋈ --> r 3 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> e = −<!-- − --> 1 P up (6 π<!-- π --> 2 s) (m de m e) (1 + 5 4 (L 1 G i) ⋈<!-- ⋈ --> r 3 2 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> de = −<!-- − --> P up (2 s 6 π<!-- π -->) (m e m de) (1 −<!-- − --> 5 C L 1) ⋈<!-- ⋈ --> r 3 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> de = S √<!-- √ --> 2 s 6 π<!-- π -->) (m n m de) (1 −<!-- − --> 1 ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 (7 s 12) ⋈<!-- ⋈ --> r 3 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> de = ⋈<!-- ⋈ --> 1 2 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 √<!-- √ --> 2 s 6 π<!-- π -->) (l p 2 q p m p t p m de) (1 −<!-- − --> 5 ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 (7 s 15) ⋈<!-- ⋈ --> r 3 ⊠<!-- ⊠ -->) 	 e = 5.24411510858423 …<!-- … --> L = 2.622057594292119 …<!-- … --> L 1 = 1.31102897714605 …<!-- … --> L 2 = 0.599070117367796 …<!-- … --> C p = 0.847213084793979 …<!-- … --> G GB = 0.834626841674073 …<!-- … --> 1st lemniscate constant 2nd lemniscate constant ubiquitous constant Gauss's constant																			
 μ<!-- μ --> s = coulomb e (l p 2 m p t p m e) (1 + √<!-- √ --> ζ<!-- ζ --> (2) ⋈<!-- ⋈ --> g 9 2 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = −<!-- − --> 1 P up (6 π<!-- π --> 2 s) (m de m e) (1 + 5 4 (L 1 G i) ⋈<!-- ⋈ --> r 3 2 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = ⋈<!-- ⋈ --> 1 2 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 √<!-- √ --> 2 s 6 π<!-- π -->) (l p 2 q p m p t p m de) (1 −<!-- − --> 5 ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 (7 s 15) ⋈<!-- ⋈ --> r 3 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = −<!-- − --> C CF F (2 s 6 π<!-- π -->) (m n m de) (1 −<!-- − --> ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 (35 L 2) ⋈<!-- ⋈ --> r 3 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> e = −<!-- − --> 1 P up (6 π<!-- π --> 2 s) (m de m e) (1 + 5 4 (L 1 G i) ⋈<!-- ⋈ --> r 3 2 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = S √<!-- √ --> 2 s 6 π<!-- π -->) (m n m de) (1 −<!-- − --> 1 ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 (7 s 12) ⋈<!-- ⋈ --> r 3 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> de = ⋈<!-- ⋈ --> 1 2 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 √<!-- √ --> 2 s 6 π<!-- π -->) (l p 2 q p m p t p m de) (1 −<!-- − --> 5 ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 (7 s 15) ⋈<!-- ⋈ --> r 3 ⊠<!-- ⊠ -->) 	 M s = 6 e ℓ<!-- ℓ --> ⋈<!-- ⋈ --> 1 2 (m p q p m p) (1 + 18 C Q ⋈<!-- ⋈ --> 1 2 + ⋈<!-- ⋈ --> 3 2 + ⋈<!-- ⋈ --> 4 2) ⊠<!-- ⊠ -->) 	 M J = 6 e ℓ<!-- ℓ --> ⋈<!-- ⋈ --> 1 2 (m n q p m p) (1 + 18 C Q ⋈<!-- ⋈ --> 1 2 + ⋈<!-- ⋈ --> 3 2 + ⋈<!-- ⋈ --> 4 2) ⊠<!-- ⊠ -->) 	 M A mass = 6 e ℓ<!-- ℓ --> ⋈<!-- ⋈ --> 1 2 (A mass q p m p) (1 + 18 C Q ⋈<!-- ⋈ --> 1 2 + ⋈<!-- ⋈ --> 3 2 + ⋈<!-- ⋈ --> 4 2) ⊠<!-- ⊠ -->) 	 M s = 6 e ℓ<!-- ℓ --> ⋈<!-- ⋈ --> 1 2 (m n q p m p) (1 + 18 C Q ⋈<!-- ⋈ --> 1 2 + ⋈<!-- ⋈ --> 3 2 + ⋈<!-- ⋈ --> 4 2) ⊠<!-- ⊠ -->) 	 M n = 6 e ℓ<!-- ℓ --> ⋈<!-- ⋈ --> 1 2 (m n q p m p) (1 + 18 C Q ⋈<!-- ⋈ --> 1 2 + ⋈<!-- ⋈ --> 3 2 + ⋈<!-- ⋈ --> 4 2) ⊠<!-- ⊠ -->) 	 M t = 6 e ℓ<!-- ℓ --> ⋈<!-- ⋈ --> 1 2 (m t q p m p) (1 + 18 C Q ⋈<!-- ⋈ --> 1 2 + ⋈<!-- ⋈ --> 3 2 + ⋈<!-- ⋈ --> 4 2) ⊠<!-- ⊠ -->) 	 M de = 6 e ℓ<!-- ℓ --> ⋈<!-- ⋈ --> 1 2 (m de q p m p) (1 + 18 C Q ⋈<!-- ⋈ --> 1 2 + ⋈<!-- ⋈ --> 3 2 + ⋈<!-- ⋈ --> 4 2) ⊠<!-- ⊠ -->) 	 M he = 6 e ℓ<!-- ℓ --> ⋈<!-- ⋈ --> 1 2 (m he q p m p) (1 + 18 C Q ⋈<!-- ⋈ --> 1 2 + ⋈<!-- ⋈ --> 3 2 + ⋈<!-- ⋈ --> 4 2) ⊠<!-- ⊠ -->) 	 M tri = 6 e ℓ<!-- ℓ --> ⋈<!-- ⋈ --> 1 2 (m tri q p m p) (1 + 18 C Q ⋈<!-- ⋈ --> 1 2 + ⋈<!-- ⋈ --> 3 2 + ⋈<!-- ⋈ --> 4 2) ⊠<!-- ⊠ -->) 	 M s = 6 e ℓ<!-- ℓ --> ⋈<!-- ⋈ --> 1 2 (m s q p m p) (1 + 18 C Q ⋈<!-- ⋈ --> 1 2 + ⋈<!-- ⋈ --> 3 2 + ⋈<!-- ⋈ --> 4 2) ⊠<!-- ⊠ -->) 	 M 12 C = 6 e ℓ<!-- ℓ --> ⋈<!-- ⋈ --> 1 2 (12 A mass q p m p) (1 + 18 C Q ⋈<!-- ⋈ --> 1 2 + ⋈<!-- ⋈ --> 3 2 + ⋈<!-- ⋈ --> 4 2) ⊠<!-- ⊠ -->) 	 N A = 6 e ℓ<!-- ℓ --> ⋈<!-- ⋈ --> 1 2 (m A q p m p) (1 + 18 C Q ⋈<!-- ⋈ --> 1 2 + ⋈<!-- ⋈ --> 3 2 + ⋈<!-- ⋈ --> 4 2) ⊠<!-- ⊠ -->) 	 T 0 = 2.731500000000000 …<!-- … --> ×<!-- × --> 10 2 kelvin p 0 = 1.000000000000000 …<!-- … --> ×<!-- × --> 10 5 pascal p 1 = 1.01325003754773 …<!-- … --> ×<!-- × --> 10 5 pascal E 1 = 6.09110229711386 …<!-- … --> ×<!-- × --> 10 −<!-- − --> 24 joules f i = 5.400000000000000 …<!-- … --> ×<!-- × --> 10 14 Hz λ<!-- λ --> peak = 4.96511423174427 …<!-- … --> ν<!-- ν --> peak = 2.82143937212207 …<!-- … --> freezing point of water 1 bar standard atmospheric pressure energy associated with ΔV_{CC}frequency of max light conversionpeak λ of spectral radiance/unit wavelengthpeak emission of spectral flux/unit frequency								
 g e = −<!-- − --> 4 ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 F up (1 + 4 C CF + 6 (2 π<!-- π -->)) ⋈<!-- ⋈ --> g 9 2 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = −<!-- − --> 2 ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 (m s m e) (1 + 4 C CF + 6 (2 π<!-- π -->)) ⋈<!-- ⋈ --> g 9 2 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = ⋈<!-- ⋈ --> 1 2 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 F up (l p 2 q p m p t p m e) (1 + C d 6 ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = ⋈<!-- ⋈ --> 1 2 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 F up (l p 2 q p m p t p m e) (1 + 14 F FF 9 s) ⋈<!-- ⋈ --> r 3 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = ⋈<!-- ⋈ --> 1 2 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 F up (l p 2 q p m p t p m e) (1 + 1 ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = −<!-- − --> 2 ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 F up (l p 2 q p m p t p m e) (1 + 4 C CF + 6 (2 π<!-- π -->)) ⋈<!-- ⋈ --> g 9 2 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = −<!-- − --> 2 ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 F up (l p 2 q p m p t p m e) (1 + 4 C CF + 6 (2 π<!-- π -->)) ⋈<!-- ⋈ --> g 9 2 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = ⋈<!-- ⋈ --> 1 2 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 F up (l p 2 q p m p t p m e) (1 + 14 F FF 9 s) ⋈<!-- ⋈ --> r 3 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = ⋈<!-- ⋈ --> 1 2 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 F up (l p 2 q p m p t p m e) (1 + 1 ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = −<!-- − --> 2 ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 F up (l p 2 q p m p t p m e) (1 + 4 C CF + 6 (2 π<!-- π -->)) ⋈<!-- ⋈ --> g 9 2 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = ⋈<!-- ⋈ --> 1 2 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 F up (l p 2 q p m p t p m e) (1 + 14 F FF 9 s) ⋈<!-- ⋈ --> r 3 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = ⋈<!-- ⋈ --> 1 2 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 F up (l p 2 q p m p t p m e) (1 + 1 ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = −<!-- − --> 2 ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 F up (l p 2 q p m p t p m e) (1 + 4 C CF + 6 (2 π<!-- π -->)) ⋈<!-- ⋈ --> g 9 2 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = ⋈<!-- ⋈ --> 1 2 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 F up (l p 2 q p m p t p m e) (1 + 14 F FF 9 s) ⋈<!-- ⋈ --> r 3 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = ⋈<!-- ⋈ --> 1 2 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 F up (l p 2 q p m p t p m e) (1 + 1 ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = −<!-- − --> 2 ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 F up (l p 2 q p m p t p m e) (1 + 4 C CF + 6 (2 π<!-- π -->)) ⋈<!-- ⋈ --> g 9 2 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = ⋈<!-- ⋈ --> 1 2 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 F up (l p 2 q p m p t p m e) (1 + 14 F FF 9 s) ⋈<!-- ⋈ --> r 3 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = ⋈<!-- ⋈ --> 1 2 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 F up (l p 2 q p m p t p m e) (1 + 1 ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = −<!-- − --> 2 ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 F up (l p 2 q p m p t p m e) (1 + 4 C CF + 6 (2 π<!-- π -->)) ⋈<!-- ⋈ --> g 9 2 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = ⋈<!-- ⋈ --> 1 2 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 F up (l p 2 q p m p t p m e) (1 + 14 F FF 9 s) ⋈<!-- ⋈ --> r 3 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = ⋈<!-- ⋈ --> 1 2 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 F up (l p 2 q p m p t p m e) (1 + 1 ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = −<!-- − --> 2 ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 F up (l p 2 q p m p t p m e) (1 + 4 C CF + 6 (2 π<!-- π -->)) ⋈<!-- ⋈ --> g 9 2 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = ⋈<!-- ⋈ --> 1 2 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 F up (l p 2 q p m p t p m e) (1 + 14 F FF 9 s) ⋈<!-- ⋈ --> r 3 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = ⋈<!-- ⋈ --> 1 2 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 F up (l p 2 q p m p t p m e) (1 + 1 ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = −<!-- − --> 2 ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 F up (l p 2 q p m p t p m e) (1 + 4 C CF + 6 (2 π<!-- π -->)) ⋈<!-- ⋈ --> g 9 2 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = ⋈<!-- ⋈ --> 1 2 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 F up (l p 2 q p m p t p m e) (1 + 14 F FF 9 s) ⋈<!-- ⋈ --> r 3 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = ⋈<!-- ⋈ --> 1 2 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 F up (l p 2 q p m p t p m e) (1 + 1 ⋈<!-- ⋈ --> 1 ⋈<!-- ⋈ --> 3 4 ⋈<!-- ⋈ --> 2 ⊠<!-- ⊠ -->) 	 μ<!-- μ --> s = −<!-- − --> 2 ⋈<!-- ⋈ --> 1 ⋈<!-- -->